Exploratory Data Analysis

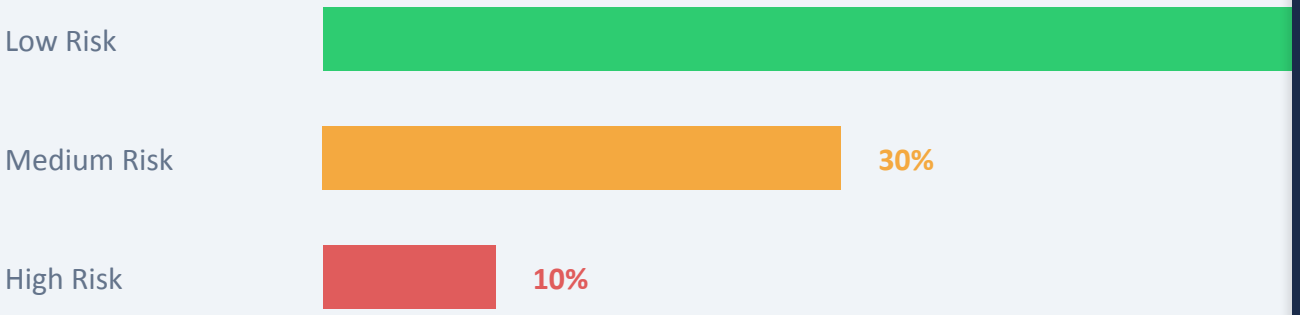
Financial Risk Dataset

A comprehensive walkthrough of data understanding, quality checks, missing values, imputation & patterns

Dataset Overview

15,000	20	6	3
Total Rows	Features	Columns w/ Missing	Risk Classes

Target: Risk Rating (imbalanced)



9,000 Low · 4,500 Medium · 1,500 High — class imbalance requires careful evaluation metrics

Purpose

1. Predict individual financial risk level
2. Understand feature relationships
3. Guide imputation & modeling choices
4. Avoid analysis mistakes

Feature Introduction · Numeric Features

Age	Age of the individual in years.
Income	Annual income. Key indicator of repayment capacity. Has 15% missing values.
Credit Score	Credit score (approx. 300–850 range). Measures creditworthiness; 15% missing.
Loan Amount	Requested or borrowed amount. 15% missing.
Debt-to-Income Ratio	Ratio of total debt to income. Decimal value; very different scale from other features.
Assets Value	Total value of individual's assets. 15% missing.
Years at Current Job	Count of years at current employer. Stability proxy — has 11 logical data quality issues.
Number of Dependents	Count of financially dependent people. Range 0–4; 15% missing.
Previous Defaults	Count of prior loan/debt defaults. Range 0–4; expected to predict risk but weak link.

Feature Introduction · Categorical Features

Gender	Male · Female · Non-binary	Nominal	*Balanced ~33% each*
Education Level	High School · Bachelor's · Master's · PhD	Ordinal	*Natural order; ~25% each*
Marital Status	Single · Married · Divorced · Widowed	Nominal	*~25% each*
Loan Purpose	Auto · Business · Home · Personal	Nominal	*~25% each*
Employment Status	Employed · Self-employed · Unemployed	Nominal	*~33% each*
Payment History	Poor · Fair · Good · Excellent	Ordinal	*~25% each; weak risk link*
Marital Status Change	0 · 1 · 2	Coded	*Treat as category, not number*
Country	243 unique countries	High-card	*Max 124 rows per country*
City / State	14,858 unique combos	Sparse	*Drop for modeling*
Risk Rating ★	Low · Medium · High	Target	*Ordinal target; imbalanced*

Data Quality & Column Issues

Marital Status Change — Unclear

Stored as 0, 1, 2 but behaves like a category. If treated as a number, a model may falsely learn that 2 > 1 > 0. Must be one-hot or label-encoded as a category. If I treat it like a normal number, the model may learn a false relationship.

Previous Defaults — Ambiguous Definition

Unclear whether it counts loan defaults, credit card defaults, or all financial defaults, and over what time period. Missing values ≠ zero defaults — they mean unknown history.

City & State — Too Sparse to Use

14,858 unique State-City combos. 14,718 cities appear only once. Zero cities have ≥ 5 rows. Cannot compute reliable city-level averages. Drop City for modeling; avoid State.

Country — High Cardinality

243 unique countries. Most common is Congo (124 rows = 0.83%). Only 64 rows labelled 'United States of America' despite many US-looking state names. One-hot encoding impractical.

Years at Current Job — Logical Error

11 rows show Age = 18 but Years at Current Job = 19. Impossible — person would have started working before birth. Not a statistical outlier, but a domain rule violation.

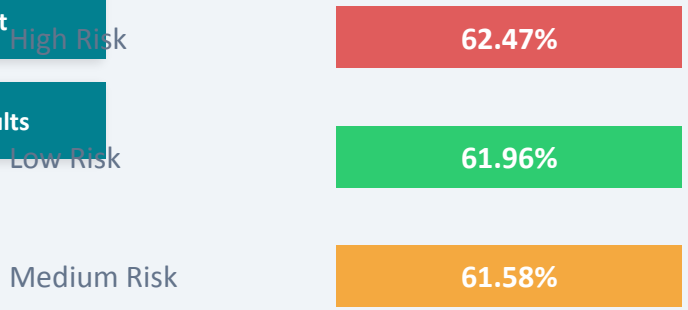
Missing Values — Overview

6	2,250	9,284	5,716
Columns with missing values	Missing per column (15%)	Rows with ≥ 1 missing value	Fully complete rows

Affected Columns

- Income
- Credit Score
- Loan Amount
- Assets Value
- Number of Dependents
- Previous Defaults

Missingness by Risk Rating Group



Key insight: Missing ≠ absent. A missing Income does not mean zero income. A missing Previous Defaults does not mean no defaults.

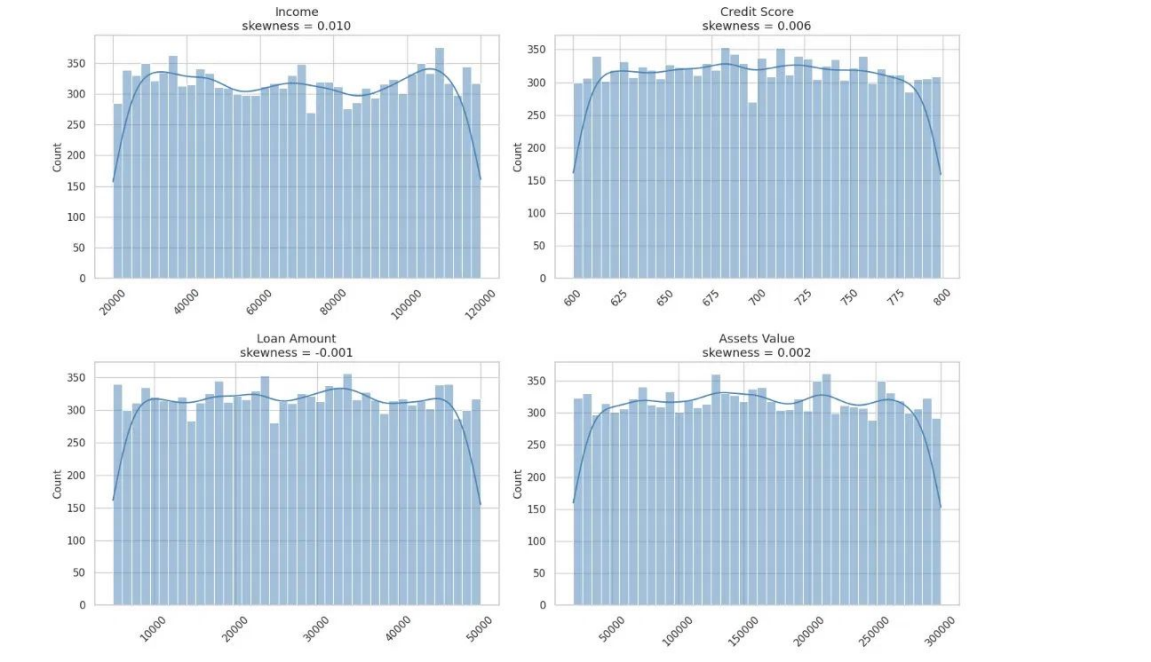
Imputation Strategy

Dropping rows with missing values would remove 9,284 rows (>60% of the dataset). That is too large a loss — imputation is necessary.

Column	Type	Technique	Rationale
Income	Continuous	Median	Distribution is symmetric; mean ≈ median; robust against extremes.
Credit Score	Continuous	Median + Flag	Mean ≈ median; missing credit info itself may signal higher risk.
Loan Amount	Continuous	Median	No strong outliers; median is safe and simple.
Assets Value	Continuous	Median	Symmetric distribution; median keeps imputation robust.
Number of Dependents	Count (0–4)	Median = 2	Uniform 0–4 distribution; missing ≠ 0 dependents.
Previous Defaults	Count (0–4)	Median = 2	Uniform 0–4 distribution; missing ≠ no defaults.

Consider adding missing-indicator columns for Credit Score, Income & Previous Defaults — missingness itself may carry predictive signal.

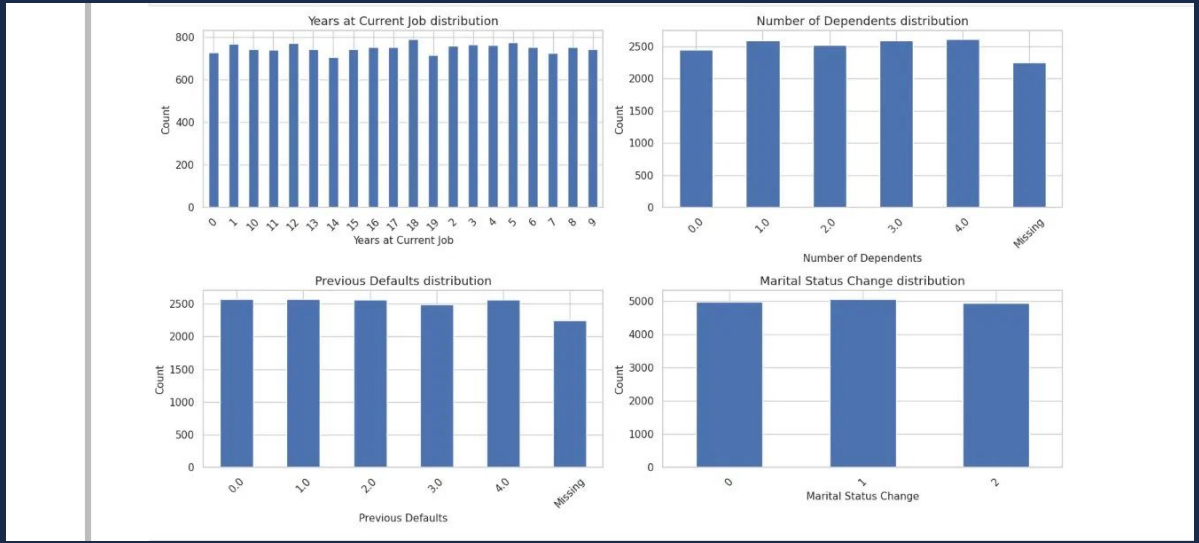
Continuous Feature Distributions



Key Findings

- No transformations needed
- Apply MinMaxScaler [0,1] for modeling — features have very different ranges. Suitable for KNN, Neural Nets, PCA

Count & Discrete Feature Distributions

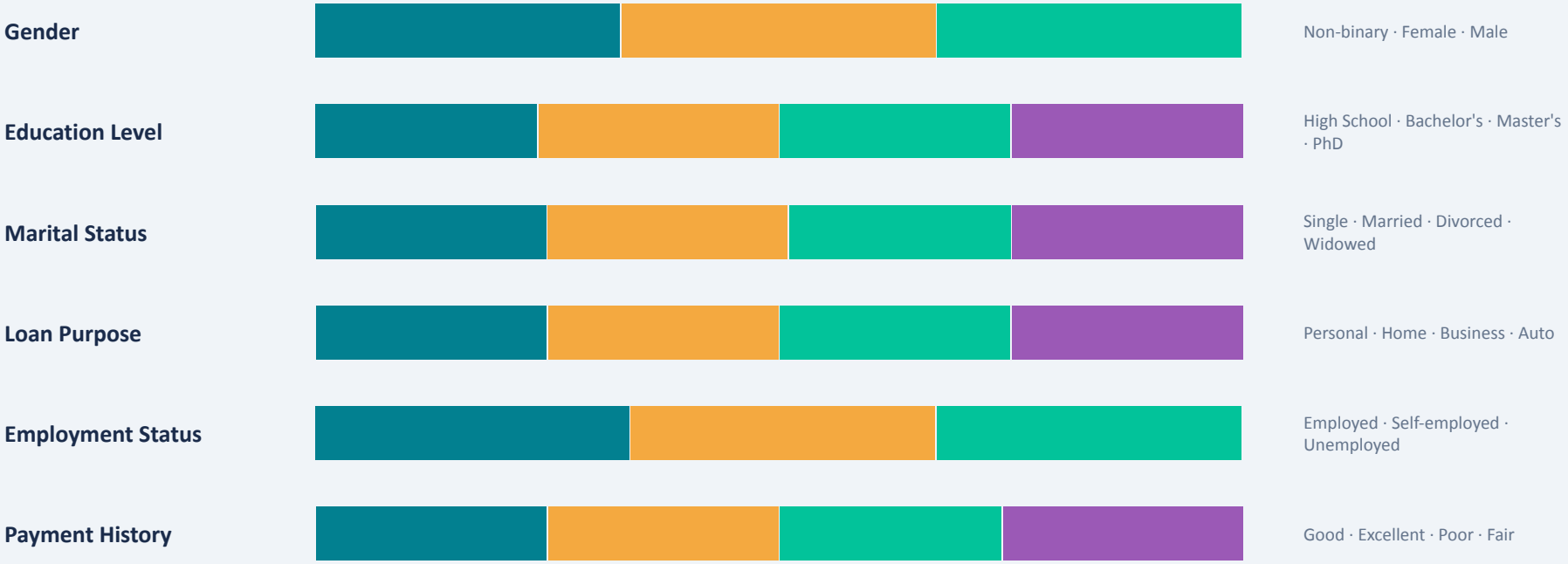


Observations

- Years at Current Job: values 0–19, roughly uniform — count feature, not continuous
- Number of Dependents: values 0–4, almost equally distributed; 'Missing' bar visible
- Previous Defaults: values 0–4, nearly flat — no value dominates
- Marital Status Change: codes 0, 1, 2 are almost equally split (~5,000 each)
- All 4 are count-like.

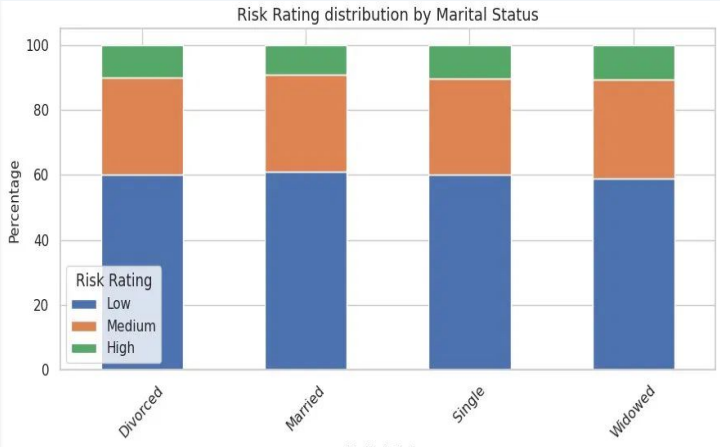
Categorical Feature Balance

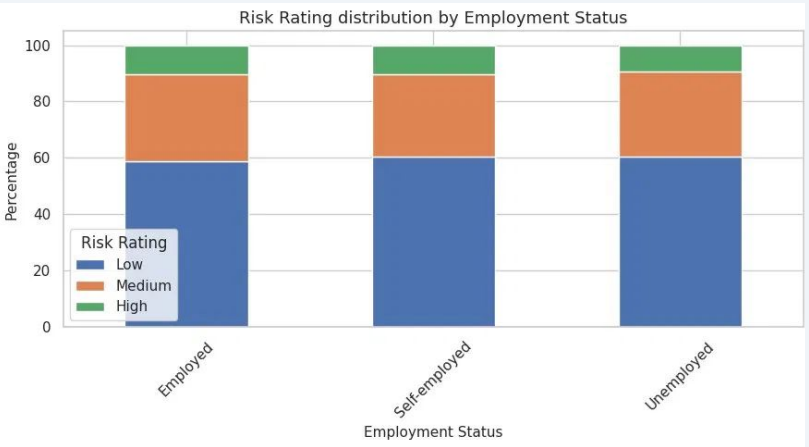
Well-Balanced Features (~25–33% per category)

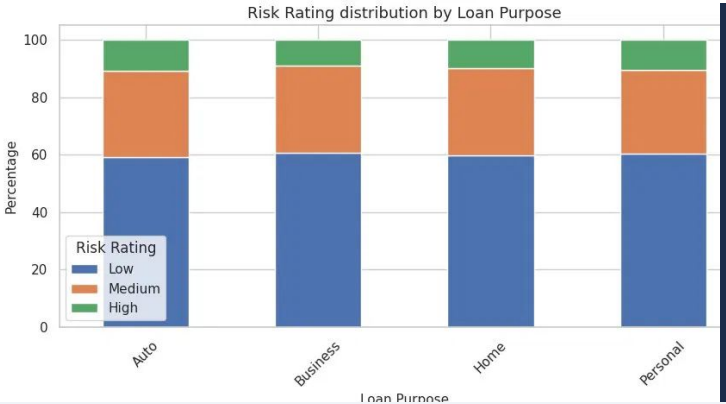


Country has 243 unique values (max 124 rows per country). City has 14,858 unique combos — almost all appearing once. Avoid direct one-hot encoding on these.

Risk Rating Distribution by Category · Part 1

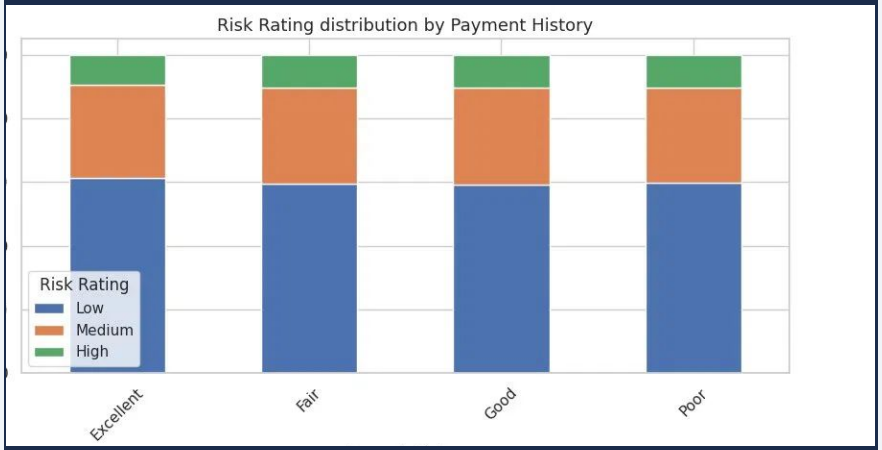


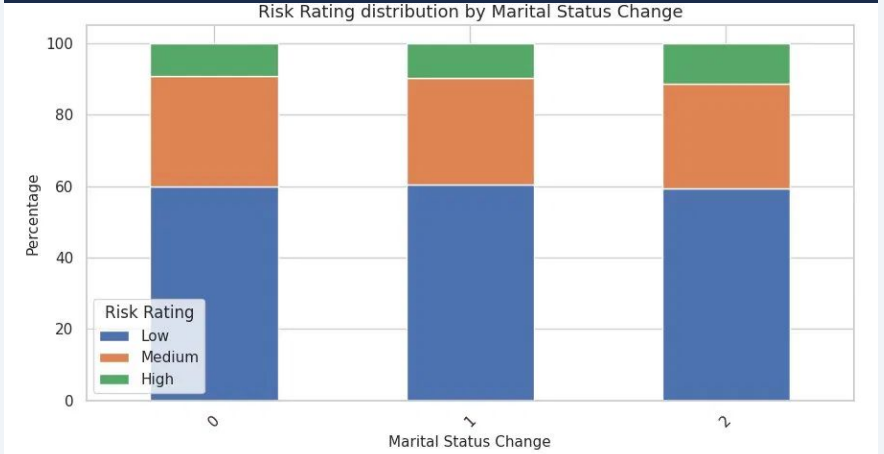




All groups show virtually identical distributions: ~60% Low · ~30% Medium · ~10% High — confirming near-zero Cramér's V for these features.

Risk Rating Distribution by Category · Part 2





Payment History

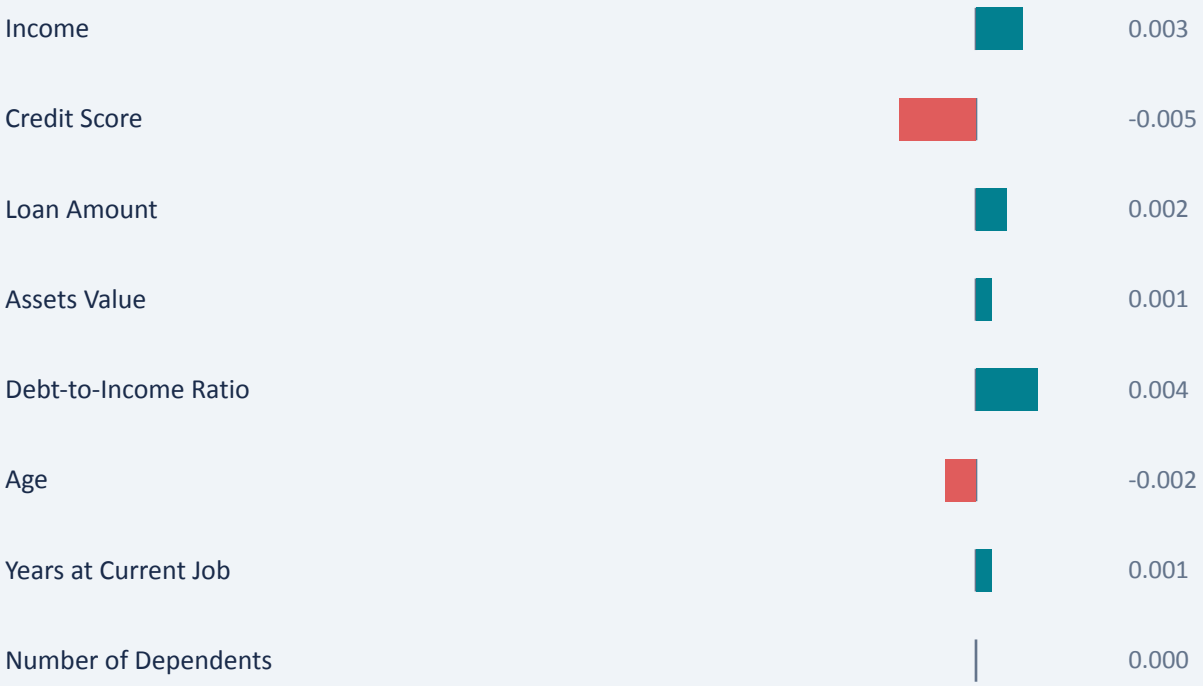
Cramér's V = 0.0000 — Excellent, Fair, Good, and Poor groups all show the same ~60/30/10 split. No risk signal at all.

Marital Status Change (codes 0, 1, 2)

All three codes show the same risk split. Confirms these are category codes, not ordered numbers with different risk profiles.

Relationship Analysis · Numeric vs Risk Rating

Correlation with Risk Rating (numeric encoding: Low=0, Medium=1, High=2)



All numeric correlations are near zero — no single numeric feature clearly separates Low, Medium, and High risk clients. Interactions between features matter more.

Relationship Analysis · Categorical vs Risk Rating

Cramér's V (0 = no association, 1 = perfect association)

Payment History	0.0000	None
Gender	0.0030	None
Education Level	0.0045	None
Marital Status	0.0028	None
Loan Purpose	0.0021	None
Employment Status	0.0019	None
Marital Status Change	0.0025	None
Previous Defaults	0.0158	Minimal

Risk distribution inside each category mirrors the full dataset (~60% Low, 30% Medium, 10% High). The dataset may be synthetic.

Domain Questions Answered

Q1: Does Payment History affect Risk Rating?

Finding: No meaningful relationship. Cramér's V = 0.0000. The risk percentages are almost identical for Excellent, Good, Fair, and Poor payment history groups — each shows roughly 60% Low, 30% Medium, 10% High risk.

Expected in real data: People with Poor payment history should show significantly higher risk. The absence of this signal is a strong indication that this dataset may be synthetic.

Q2: Does Previous Defaults affect Risk Rating?

Finding: Very weak relationship. Cramér's V = 0.0158. The High-risk percentage changes only slightly between groups with 0, 1, 2, 3, or 4 previous defaults.

Expected in real data: Previous defaults should be one of the strongest predictors of financial risk. Its near-zero Cramér's V suggests the target variable was assigned independently of realistic financial logic.

Outliers & Domain Rule Checks

IQR Method → No statistical outliers found

Z-Score (±3σ) → No statistical outliers found

Modified Z-Score → No statistical outliers found

Basic Domain Rules Check

✓ No negative income, credit score, loan amounts, or invalid debt-to-income ratios found.

Logical Data Quality Issue

Years at Current Job

Age = 18
Years at Job = 19

11 rows affected. A person cannot have worked 19 years when they are only 18 years old.

Conclusion: Numeric values are well-controlled statistically. Real issues hide in logical domain rules, not in extreme values. Always combine statistical with domain checks.

Patterns & Key Findings

01 No strong single predictor

No feature — numeric or categorical — clearly separates Low, Medium, or High risk clients. All correlations and Cramér's V values are near zero.

02 All categorical features are balanced

Gender, Education Level, Marital Status, Loan Purpose, Employment Status, and Payment History each have ~25–33% per category.

03 Missing values are non-structural

Missing values appear evenly across all risk groups (~62% each). They do not cluster in one group, suggesting random or administrative missingness.

04 City/State data is unreliable

Nearly all city-state combinations appear only once in 15,000 rows. Geographic analysis at city level is impossible. Drop City for modeling.

05 Likely synthetic dataset

The complete absence of expected financial relationships (payment history vs. risk, defaults vs. risk) strongly suggests the target was randomly assigned.

06 Ordinal features need proper encoding

Education Level and Payment History have natural order. Encoding them as unordered categories loses information. Use ordinal encoding.

Potential Mistakes to Avoid

1 Treating Marital Status Change as a continuous number

Columns coded 0/1/2 are not numbers. Using them as-is implies 2 > 1 > 0. One-hot encode or treat as category.

2 One-hot encoding City (10,000+ unique values)

This creates thousands of near-empty columns, adds noise, and increases dimensionality without useful signal. Drop City entirely.

3 Ignoring class imbalance in Risk Rating

60% Low / 30% Medium / 10% High. A naive model predicting 'Low' for everyone scores 60% accuracy. Use F1, Precision, Recall.

4 Filling missing values with 0 or 'None'

Missing Income ≠ zero income. Missing Previous Defaults ≠ no defaults. Use median imputation and missing-indicator flags.

5 Dropping all rows with missing values

Over 9,284 rows have at least one missing value. Dropping them removes 62% of the dataset and introduces selection bias.

6 Relying on accuracy alone for model evaluation

Due to imbalance, accuracy is misleading. A model can look good by always predicting Low and completely miss High-risk clients.

EDA Summary

What We Learned

- 20 features explored: numeric, ordinal, nominal, coded, and count-like
- 15% missingness in 6 columns → median imputation (no zeros, no drops)
- No strong single predictor of Risk Rating — feature interactions are key
- Dataset likely synthetic: payment history and defaults show no risk signal
- Location data (City/State) too sparse; requires special handling for Country
- Use tree-based models + F1/Recall/Confusion Matrix for modeling

EDA Summary

Questions?

Thank you!